⑤ For a good conductor attenuation constant

$\alpha = \sqrt{\omega\mu\sigma/2}$.

phase constant $\beta = \sqrt{\frac{\omega\mu\sigma}{2}}$

Intrensic impedance $\eta = \sqrt{\frac{\omega\mu}{\sigma}}$ & it must be less than $45\,\Omega$.

← ⑥ conductivity.

① Determine the highest frequency above which Earth can be considered as good dielectric ~~for~~/insulator. for a ratio of $\frac{\sigma}{\omega\varepsilon_0}$ less than 0.1. Assume $\varepsilon_r = 2$ & $\mu = \mu_0$

$\sigma = 1000$ simens/m

ε_r [illegible]

(Note:)

⑥ In a free space loss less medium conductivity = 0,

$\alpha = 0$

$\beta = \omega\sqrt{\mu\varepsilon_r}$

Given

$$\frac{\sigma}{\omega\varepsilon_0} < 0.1$$

$$\frac{\sigma}{2\pi f\varepsilon_0\varepsilon_r} = 0.1$$

$$f = \frac{\sigma}{2\pi\varepsilon_0\varepsilon_r \times 0.1}$$

$$= \frac{1000}{2\times\pi\times 8.854\times 10^{-12}\times 2\times 0.1}$$

$$= 89.87\times 10^{12} < f$$

② Determine the highest frequency above which copper acts as a dielectric for a loss tangent < 0.2. Assume conductivity (σ) 5×10^{7} simens/m.

Given,

$$\frac{\sigma}{\omega \varepsilon_0} < 0.2$$

$$\frac{5 \times 10^7}{2 \times \pi \times f \times \varepsilon_0 \times \varepsilon_r} < 0.2$$

$$f < \frac{5 \times 10^7}{2 \times \pi \times 8.854 \times 10^{-12} \times 0.2}$$

$$f = 4.49 \times 10^{18} \, Hz$$

Above this frequency Cu acts as a dielectric

(03) A signal of 1 MHz and 1A of conductor current flows through a copper cabel. Determine the value of displacement current density in the Cu cable. Assume $\sigma = 5 \times 10^7$ simens/m.

∴ WKT, $f = 1 MHz$

$I_C = 1 A$

$I_D = ?$

$\sigma = 5 \times 10^7 \, S/m.$

loss tangent $= \frac{J_C}{J_D}$.

$$\frac{\sigma}{\omega \varepsilon} = \frac{J_C}{J_D}$$

$$\frac{\sigma}{\omega \varepsilon_0 \varepsilon_r} = \frac{J_C}{J_D}$$

$$J_D = \frac{J_C \times 2\pi f \times \varepsilon_0 \times \varepsilon_r}{\sigma}$$

$$= \frac{1 \times 2 \times \pi \times 1 \times 10^6 \times \overset{8.854\times10^{-12}}{\cancel{5\times10^{-7}}} \times 1}{5 \times 10^{7}}$$

$$J_D = 1.11 \times 10^{-12} A$$

$$J_D = 1.11 \, pA$$

(09) Determine attenuation constant, phase constant, propogation constant, wave length, phase velocity and intrensic impedance for damp soil at a frequency of 1 MHz. Given $\varepsilon_r = 12$ and conductivity, $\sigma = 20 \, S/m$

Given,

$\varepsilon_r = 12$

$f = 1 \times 10^6 \, Hz$

$\sigma = 20 \, S/m \times 10^{-3} \, S/m.$

AC $\alpha = ?$

$\beta = ?$

PC $\gamma = ?$

$\lambda = ?$

$V = ?$

$\eta = ?$

$$\frac{\sigma}{\omega\varepsilon} >>> 1$$

$$\frac{20 \times 10^{-3}}{2\pi \times 1 \times 10^6 \times \varepsilon_0 \times \varepsilon_r} >> 1$$

$$\frac{20 \times 10^{-3}}{2\pi \times 10^6 \times 8.854 \times 10^{-12} \times 12} >> 1$$

$$29.97 >> 1$$

$\therefore$ damp soil could be good conductor.

$\mu_0 = 4\pi \times 10^{-7}$

μ_r not given assume 1.

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$= \sqrt{\frac{2\pi f \times \mu_0 \times \mu_r \times \sigma}{2}}$$

$$= \sqrt{\frac{2 \times \pi \times 10^6 \times 4 \times \pi \times 10^{-7} \times 1 \times 20 \times 10^{-3}}{2}}$$

$$\boxed{\alpha = \beta = 0.281}$$

$$\gamma = \alpha + j\beta$$

$$\boxed{\gamma = 0.281 + j0.281}$$

$$V = \frac{\omega}{\beta}$$

$$= \frac{2 \times \pi \times 1 \times 10^6}{0.281}$$

$$\boxed{V = 22.36 \text{ M m/s}}$$

$$\lambda = \frac{V}{f}$$

$$= \frac{22.36 \times 10^6}{1 \times 10^6}$$

$$\boxed{\lambda = 22.36 \text{ m}}$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}$$

$$= \sqrt{\frac{2\pi f \times \mu_0 \mu_r}{\sigma}} = \sqrt{\frac{2 \times \pi \times 10^6 \times 4 \times \pi \times 10^{-7} \times 1}{20 \times 10^{-3}}}$$

$$\boxed{\eta = 19.86\ \Omega}$$

In a lossless medium for which intrinsic impedance = 60π, $\mu_r = 1$ and $H = -0.1\cos(\omega t - z)\hat{a}_x + 0.5\sin(\omega t - z)\hat{a}_y$ A/m. Calculate ε_r, ω and E.

Given,

$\eta = 60\pi$

$\mu_r = 1$

$H = -0.1\cos(\omega t - z)\hat{a}_x + 0.5\sin(\omega t - z)\hat{a}_y$

$\varepsilon_r = ?$, $\omega = ?$, $E = ?$

$\mu = \mu_r \mu_0$.

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

$$60\pi = \sqrt{\frac{\mu_r \mu_0}{\varepsilon_0 \varepsilon_r}}$$

Squaring on both side

$$(60\pi)^2 = \frac{\mu_r^2 \mu_0}{\varepsilon_0 \varepsilon_r}$$

$$\varepsilon_r = \frac{4\pi \times 10^{-7} \times 1}{(60 \times \pi)^2 \times 8.854 \times 10^{-12}}$$

$$\boxed{\varepsilon_r = 3.99} \approx 4.$$

ω

$$\beta = \omega\sqrt{\mu\varepsilon}$$

$$\beta = \omega\sqrt{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}$$

Squaring on both side

$$\beta^2 = \omega^2 \mu_0 \mu_r \varepsilon_0 \varepsilon_r.$$

$$\omega^2 = \frac{\beta^2}{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}$$

(7) A uniform plane wave propogating in a medium as $E = 2 \times e^{-\alpha z} \sin(10^8 t - \beta z)\, \hat{a}_y$ V/m. If the medium is characterized by $\varepsilon_r = 1$, ~~$\mu_r = 20$~~. $\mu_r = 20$, $\sigma = 3$ S/m. Find attenuation constant, α, β, γ, V, λ, η.

Given,

$E = 2e^{-\alpha z} \sin(10^8 t - \beta z)\, \hat{a}_y$

$\varepsilon_r = 1$

$\mu_r = 20$

$\sigma = 3$ S/m

$\alpha = \beta = \gamma = V = \lambda = \eta = ?$

when it is propogating in a medium then e power comes

$\omega = 10^8$

$$\alpha = \beta = \sqrt{\frac{\omega \mu \sigma}{2}}$$

$$= \sqrt{\frac{\omega \cdot \mu_0 \mu_r \sigma}{2}}$$

$$\frac{\sigma}{\omega \varepsilon} \gg 1$$

$$\frac{3}{10^8 \times \varepsilon_0 \times \varepsilon_r} \gg 1$$

$$\frac{3}{10^8 \times 8.85 \times 10^{-12} \times 1} \gg 1$$

$$3.38 \times 10^8 \gg 1$$

$\therefore$ for ~~It~~ a is good conductor, $\alpha = \beta = \sqrt{\frac{\omega \mu \sigma}{2}}$

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$= \sqrt{\frac{\omega\mu_0\mu_r\sigma}{2}}$$

$$= \sqrt{\frac{10^8 \times 4\pi \times 10^{-7} \times 20 \times 3}{2}}$$

$$\boxed{\alpha = \beta = 61.4.}$$

$$\gamma = \alpha + j\beta$$

$$\boxed{\gamma = 61.4 + j61.4}$$

$$V = \frac{\omega}{\beta}$$

$$= \frac{10^8}{61.4}$$

$$\boxed{V = 1.62 \text{ M m/s}}$$

$$\lambda = \frac{V}{f}$$

$$= \frac{1.62 \times \cancel{10^6}}{16 \times \cancel{10^6}}$$

$$\boxed{\lambda = 0.101 \text{ m}}$$

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi} = \frac{10^8}{2\pi}$$

$$\boxed{f = 16 \text{ MHz}}$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}$$

$$= \sqrt{\frac{\omega\mu_0\mu_r}{\sigma}} = \sqrt{\frac{10^8 \times 4\pi \times 10^{-7} \times 20}{3}}$$

$$\boxed{\eta = 28.94}$$